Glossary of Terms: Bimodal Manifold Interaction (BMI)

I. Fundamental Architecture

- **Bimodal Manifold Interaction (BMI):** The theoretical framework positing that the universe is a 6D bulk manifold created by the collision of two membranes (branes σ_A and σ_B). All physical phenomena are derivations of the geometric relaxation of this collision.
- **Thick Brane Interface / Manifold Tension Scalar (Σ):** The 4D spacetime hypersurface of finite thickness δ where our observable reality exists. It acts as the "crease" or boundary between the colliding branes. Its intrinsic scalar tension limits the isotropy of gravitational potentials.
- **Bridge Stress (τ):** The elastic tension and hyperspace momentum of the connection between the colliding branes. It is the primary component of gravitational field energy.
- **Inter-brane Potential ($V_{\mathbf{gap}}$):** The vacuum energy inherent to the spatial gap separating the colliding branes. It acts as the driver for global cosmological expansion (Dark Energy).
- **Universal Equilibrium (Ξ):** The total net spacetime curvature variance of the manifold, defining the baseline stability of the 4D interface against the bulk. It dictates the expected variance of gravitational signals.

II. Particle & Topological Mechanics

- **Harmonic Node:** A discrete, stable configuration of localized energy within the interface. These are the BMI framework's equivalent to "particles."
- **Nested Torus (T^3):** The geometric topology of a harmonic node. Matter is not a point-particle but a stable, multi-layered phase-lock of toroidal membranes within the interface.
- **Compactified Crease Wake (Stable Soliton):** The localized topological deformation trailing a propagating particle, maintaining its structural integrity and ensuring stable geodesic motion without dispersive decay.
- **Winding Matrix ($\mathbf{R_{top}}$):** A mathematical operator that maps how complex nested geometries (e.g., meta-stable leptons) relax into lower-energy configurations, governing phenomena like radioactive decay.
- **Manifold Impedance Matrix ($\mathbf{Z}_{ij}$):** A complex tensor representing the background elasticity of the manifold. It determines the strength and range of forces between harmonic nodes.
- **Topological Pinning:** The causal, mechanical process wherein an expanding wave function localizes into a singular spatial coordinate due to the effective coupling constant reaching unity ($g_{\text{eff}} \to 1$).
- **Topological Breaking Limit ($\Omega_{\mathbf{max}}$):** The absolute energy threshold where local energy density exceeds the elastic modulus of the background vacuum.

III. Forces & Gauge Interactions

- **Screening Gate Function ($g_{\mathbf{eff}}$):** A dynamic, non-linear function that modulates interface permeability. It determines whether a field propagates as a diffuse wave or "pins" to a target based on local Ricci curvature (R).
- **Hyperspatial Shear Tensor ($S_{\mu\nu}$):** The tensor defining the structural alignment of the interface crease relative to the compactified shadow manifold. It dictates the directionality and momentum of particle paths (P^{μ}).
- **Geometric Clamping:** The mechanism of mass generation in the BMI framework. Mass arises not from a scalar field, but from the geometric resistance encountered when a gauge field attempts to twist perpendicular to the interface boundaries.
- **Shadow Manifold ($M_{\mathbf{shadow}}$):** The compactified, non-observable region of the 6D bulk. It projects a "Tidal Shadow Potential" (Φ_{shadow}) that flattens galactic rotation curves, removing the need for particle dark matter.

IV. Observational Metrics & Validation

- **Effective Chirp Mass (ECM):** A dynamic metric extracted via STFT spectral decomposition of high-frequency gravitational waves, used to isolate scalar excitation modes from primary tensor merger events.
- **Phase-Lag Metric (Δt):** The temporal divergence ($\Delta t = t_{\phi} - t_h$) between the arrival of the scalar excitation (t_{ϕ}) and the primary tensor wave (t_h), proportional to the curvature threshold of the manifold bridge.
- **Kaluza-Klein Compactification Limit (R_c):** The calculable radius of the hidden extra dimensions, calibrated to the weak-force gauge scale (approx. 10^{-15} m) based on observed interferometer frequency shifts.

V. Methodology

- **Atomic File Overwrite:** The rigorous protocol of replacing entire file trees during development rather than fragmented edits, preventing "State Drift" in human-AI collaboration.
- **State Drift:** The tendency for complex systems to diverge from their original physical definitions over long periods of iteration; neutralized in BMI by the Atomic File Overwrite methodology.